Chapter 9. An Introduction to Elliptic Functions

Doubly periodic functions, Liouville's theorems, the Weierstrass $\wp$ function, Eisenstein series

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Overview

Elliptic functions grew out of the study of *elliptic integrals* $\int R(x, \sqrt{P(x)})\,dx$, R rational and P a polynomial of degree three or four (degree two reduces to the circular functions), which arose in computing the arc length of an ellipse or lemniscate. Their early study centered on transformation properties and an inherent double periodicity (cf. the map of $\mathbb{H}$ onto a rectangle, Section 4). Jacobi began the systematic study of doubly periodic functions (*elliptic functions*), with theta functions playing a decisive role; Weierstrass's approach, based on his function $\wp$, is simpler in its first steps, and we sketch it here, reaching the *Eisenstein series* and their expression via divisor functions, a link with number theory. Theta functions and further number-theoretic applications follow in Chapter 10.

1 Elliptic functions

We consider meromorphic f on $\mathbb{C}$ with two periods: non-zero ω_1, ω_2 with $f(z+\omega_1) = f(z)$ and $f(z+\omega_2) = f(z)$ for all z; f is **doubly periodic**. If $\omega_2/\omega_1 \in \mathbb{R}$, f is either simply periodic (quotient rational) or constant (quotient irrational), so we assume ω_1, ω_2 are linearly independent over $\mathbb{R}$. *Normalization*: let $\tau = \omega_2/\omega_1$; since τ and $1/\tau$ have imaginary parts of opposite signs, after possibly interchanging ω_1, ω_2 we may assume $\mathrm{Im}(\tau) > 0$. f has periods ω_1, ω_2 iff $F(z) = f(\omega_1 z)$ has periods 1 and τ, and F is meromorphic iff f is. So we assume f is meromorphic on $\mathbb{C}$ with periods 1 and τ, $\mathrm{Im}(\tau) > 0$. Then

$$f(z+n+m\tau) = f(z) \qquad \text{for all integers } n, m \text{ and all } z \in \mathbb{C}, \tag{1}$$

and we consider the **lattice** $\Lambda = \{n+m\tau : n, m \in \mathbb{Z}\}$ **generated** by 1 and τ, and the **fundamental parallelogram**

$$P_0 = \{z \in \mathbb{C} : z = a + b\tau,\ 0 \le a < 1,\ 0 \le b < 1\}.$$

z and w are **congruent modulo** Λ, $z \sim w$, if $z - w \in \Lambda$; by (1), $f(z) = f(w)$ when $z \sim w$. Every $z \in \mathbb{C}$ is congruent to a unique point of P_0: write $z = a + b\tau$ with $a, b \in \mathbb{R}$ (1 and τ form a real basis of $\mathbb{C}$), let n, m be the greatest integers $\le a$, $\le b$, and $w = z - n - m\tau = (a-n) + (b-m)\tau \in P_0$. If $w = a + b\tau$ and $w' = a' + b'\tau$ in P_0 are congruent, then $a - a'$, $b - b'$ are integers in $(-1, 1)$, hence 0. A **period parallelogram** is any translate $P = P_0 + h$; applying the above to $z - h$, every point is congruent to a unique point of P. Moreover

$$\mathbb{C} = \bigcup_{n,m \in \mathbb{Z}} (n + m\tau + P_0), \tag{2}$$

a disjoint union.

Proposition 1.1

Suppose f is a meromorphic function with two periods 1 and τ which generate the lattice Λ. Then:
(i) Every point in $\mathbb{C}$ is congruent to a unique point in the fundamental parallelogram.
(ii) Every point in $\mathbb{C}$ is congruent to a unique point in any given period parallelogram.
(iii) The lattice Λ provides a disjoint covering of the complex plane, in the sense of (2).
(iv) The function f is completely determined by its values in any period parallelogram.

1.1 Liouville's theorems

Theorem 1.2

An entire doubly periodic function is constant.

Proof. It is determined by its values on P_0, whose closure is compact; so it is bounded on $\mathbb{C}$, hence constant by Liouville's theorem. ■

A non-constant doubly periodic meromorphic function is an **elliptic function**. It has finitely many zeros and poles in any period parallelogram (possibly some on ∂P_0); we count them with multiplicities.

Theorem 1.3

The total number of poles of an elliptic function in P_0 is always ≥ 2.

So f cannot have just one simple pole (a single double pole is allowed).

Proof. Suppose first f has no poles on ∂P_0. By the residue theorem $\int_{\partial P_0} f(z)\,dz = 2\pi i \sum \operatorname{res} f$, and the integral is 0 by periodicity:

$$\int_{\partial P_0} f = \int_0^1 f + \int_1^{1+\tau} f + \int_{1+\tau}^{\tau} f + \int_{\tau}^0 f,$$

and opposite sides cancel; e.g.

$$\int_0^1 f(z)\,dz + \int_{1+\tau}^{\tau} f(z)\,dz = \int_0^1 f(z)\,dz + \int_1^0 f(s+\tau)\,ds = \int_0^1 f(z)\,dz - \int_0^1 f(s)\,ds = 0.$$

So $\sum \operatorname{res} f = 0$, and f (having a pole) must have at least two poles in P_0. If f has a pole on ∂P_0, choose a small h with no poles on $\partial(h + P_0)$ and argue with $P = h + P_0$. ■

The total number of poles (with multiplicities) of an elliptic function is its **order**.

Theorem 1.4

Every elliptic function of order m has m zeros in P_0.

Proof. If f has no zeros or poles on ∂P_0, the argument principle gives $\int_{\partial P_0} \frac{f'}{f} = 2\pi i(N_z - N_p)$, and by periodicity (as above) the integral is 0, so $N_z = N_p$. Otherwise apply the argument to a translate of P_0. ■

Consequently, for every $c \in \mathbb{C}$, $f(z) = c$ has as many solutions in P_0 as the order of f ($f - c$ is elliptic with the same poles).

1.2 The Weierstrass $\wp$ function

An elliptic function of order two. We construct an elliptic function whose only singularities are double poles at the lattice points. For one period, $\sum_{n=-\infty}^{\infty} \frac{1}{z+n}$ is unchanged under $z \mapsto z+1$ with poles at the integers but is not absolutely convergent; summing symmetrically,

$$F(z) = \lim_{N\to\infty} \sum_{|n|\leq N} \frac{1}{z+n} = \frac{1}{z} + \sum_{n=1}^{\infty} \Big[\frac{1}{z+n} + \frac{1}{z-n}\Big],$$

the bracket being $O(1/n^2)$; $F(z) = \pi \cot \pi z$ (Chapter 5). Alternatively, write it as $\frac{1}{z} + \sum_{n\neq 0} \big[\frac{1}{z+n} - \frac{1}{n}\big]$, absolutely convergent as $\frac{1}{z+n} - \frac{1}{n} = O(1/n^2)$, and equal to the former since $\big[\frac{1}{z+n} - \frac{1}{n}\big] + \big[\frac{1}{z-n} -$

$\frac{1}{-n}] = \frac{1}{z+n} + \frac{1}{z-n}$. Mimicking the second way: $\sum_{\omega\in\Lambda} \frac{1}{(z+\omega)^2}$ does not converge absolutely, so with $\Lambda^* = \Lambda - \{0\}$ consider

$$\frac{1}{z^2} + \sum_{\omega\in\Lambda^*} \Big[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2}\Big], \qquad \frac{1}{(z+\omega)^2} - \frac{1}{\omega^2} = \frac{-z^2-2z\omega}{(z+\omega)^2\omega^2} = O\Big(\frac{1}{|\omega|^3}\Big) \text{ as } |\omega|\to\infty.$$

Lemma 1.5

The two series

$$\sum_{(n,m)\neq(0,0)} \frac{1}{(|n|+|m|)^r} \qquad \text{and} \qquad \sum_{n+m\tau\in\Lambda^*} \frac{1}{|n+m\tau|^r}$$

converge if $r > 2$.

(By the Note at the end of Chapter 7, absolute convergence of a double series does not depend on the order of summation; we sum first in m, then in n.)

Proof. For $n \neq 0$, by integral comparison ($1/k^r \le \int_{k-1}^k dx/x^r$),

$$\sum_{m\in\mathbb{Z}} \frac{1}{(|n|+|m|)^r} = \frac{1}{|n|^r} + 2\sum_{k\ge|n|+1} \frac{1}{k^r} \le \frac{1}{|n|^r} + 2\int_{|n|}^\infty \frac{dx}{x^r} \le \frac{1}{|n|^r} + \frac{C}{|n|^{r-1}},$$

so for $r > 2$, $\sum_{(n,m)\neq(0,0)} \frac{1}{(|n|+|m|)^r} = \sum_{|m|\neq 0} \frac{1}{|m|^r} + \sum_{|n|\neq 0}\sum_m \frac{1}{(|n|+|m|)^r} < \infty$. For the second series it suffices that $|n|+|m| \le c|n+m\tau|$. Write $A \approx B$ if $A \lesssim B$ and $B \lesssim A$ ($x \lesssim y$ meaning $x \le ay$ for some $a > 0$). For positive A, B, $(A^2+B^2)^{1/2} \approx A + B$ (squaring gives one direction, and $A, B \le (A^2+B^2)^{1/2}$ the other). With $\tau = s + it$, $t > 0$,

$$|n+m\tau| = [(n+ms)^2 + (mt)^2]^{1/2} \approx |n+ms| + |mt| \approx |n+ms| + |m|,$$

and $|n+ms| + |m| \approx |n| + |m|$, considering separately $|n| \ge 2|m||s|$ and $|n| < 2|m||s|$. ∎

Remark

The proof shows that for $r > 2$ the series $\sum|n+m\tau|^{-r}$ converges uniformly in every half-plane $\mathrm{Im}(\tau) \ge \delta > 0$. For $r = 2$ it diverges.

The **Weierstrass $\wp$ function** is

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega\in\Lambda^*} \Big[\frac{1}{(z+\omega)^2} - \frac{1}{\omega^2}\Big] = \frac{1}{z^2} + \sum_{(n,m)\neq(0,0)} \Big[\frac{1}{(z+n+m\tau)^2} - \frac{1}{(n+m\tau)^2}\Big].$$

It is meromorphic with double poles at the lattice points: for $|z| < R$ split the sum into $|\omega| \le 2R$ and $|\omega| > 2R$; the terms of the second sum are $O(1/|\omega|^3)$ uniformly, so by Lemma 1.5 it is holomorphic in $|z| < R$, and the finite first sum exhibits the double poles.

Theorem 1.6

The function $\wp$ is an elliptic function that has periods 1 and τ, and double poles at the lattice points.

Proof. It remains to prove periodicity. Differentiating termwise,

$$\wp'(z) = -2\sum_{n,m\in\mathbb{Z}} \frac{1}{(z+n+m\tau)^3},$$

which converges absolutely for $z \notin \Lambda$ ($r = 3$ in Lemma 1.5) and is clearly periodic with periods 1, τ, the terms $1/\omega^2$ having disappeared. So $\wp(z+1) = \wp(z) + a$ and $\wp(z+\tau) = \wp(z) + b$. $\wp$ is even (replace the sum over ω by the sum over $-\omega$), so $\wp(-1/2) = \wp(1/2)$ and $\wp(-\tau/2) = \wp(\tau/2)$; setting $z = -1/2$ and $z = -\tau/2$ gives $a = b = 0$. ∎

Properties of $\wp$. $\wp$ is even, so $\wp'$ is odd; with periodicity,

$$\wp'(1/2) = \wp'(\tau/2) = \wp'\Big(\frac{1+\tau}{2}\Big) = 0,$$

e.g. $\wp'(1/2) = -\wp'(-1/2) = -\wp'(-1/2+1) = -\wp'(1/2)$. Since $\wp'$ is elliptic of order 3, the **half-periods** $1/2$, $\tau/2$, $(1+\tau)/2$ are its only roots in P_0, each of multiplicity 1. Let

$$\wp(1/2) = e_1, \qquad \wp(\tau/2) = e_2, \qquad \wp\Big(\frac{1+\tau}{2}\Big) = e_3.$$

Then $\wp(z) = e_1$ has a double root at $1/2$, and since $\wp$ has order 2, no other solutions in P_0; similarly for e_2, e_3. In particular e_1, e_2, e_3 are distinct (else $\wp$ would take a value four times).

Theorem 1.7

The function $(\wp')^2$ is the cubic polynomial in $\wp$

$$(\wp')^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3).$$

Proof. $F(z) = (\wp(z)-e_1)(\wp(z)-e_2)(\wp(z)-e_3)$ has, in P_0, only double roots at $1/2$, $\tau/2$, $(1+\tau)/2$; so does $(\wp')^2$. Both have poles of order 6 at the lattice points. So $(\wp')^2/F$ is holomorphic and doubly periodic, hence constant. Near 0, $\wp(z) = \frac{1}{z^2} + \cdots$ and $\wp'(z) = \frac{-2}{z^3} + \cdots$, so the constant is 4. ∎

$\wp$ is universal:

Theorem 1.8

Every elliptic function f with periods 1 and τ is a rational function of $\wp$ and $\wp'$.

Lemma 1.9

Every even elliptic function F with periods 1 and τ is a rational function of $\wp$.

Proof. A zero or pole of the even function F at 0 has even order, so for some integer m, $\wp^m F$ has no zero or pole at the lattice points; we may assume F itself has none. We construct a doubly periodic G with the same zeros and poles as F, recalling that $\wp(z)-\wp(a)$ has a single zero of order 2 if a is a half-period, and two distinct zeros a, $-a$ otherwise. If a is a zero of F, so is $-a$ (F is even); $a \sim -a$ iff a is a half-period, in which case the zero has even order. So if $a_1, -a_1, \ldots, a_m, -a_m$ (with multiplicities; for a half-period a_j, each of a_j, $-a_j$ carries half the multiplicity) are all the zeros of F in a period parallelogram, $[\wp(z) - \wp(a_1)]\cdots[\wp(z) - \wp(a_m)]$ has precisely the same roots as F. Similarly with the poles $b_1, -b_1, \ldots, b_m, -b_m$,

$$G(z) = \frac{[\wp(z) - \wp(a_1)]\cdots[\wp(z) - \wp(a_m)]}{[\wp(z) - \wp(b_1)]\cdots[\wp(z) - \wp(b_m)]}$$

is periodic with the same zeros and poles as F. So F/G is holomorphic and doubly periodic, hence constant. ∎

Proof of Theorem 1.8. Write $f = f_{\text{even}} + f_{\text{odd}}$ with $f_{\text{even}}(z) = \frac{f(z)+f(-z)}{2}$, $f_{\text{odd}}(z) = \frac{f(z)-f(-z)}{2}$. Since $\wp'$ is odd, $f_{\text{odd}}/\wp'$ is even; apply the lemma to f_{even} and $f_{\text{odd}}/\wp'$. ∎

2 The modular character of elliptic functions and Eisenstein series

We now study the dependence on τ and write $\wp_\tau$. Since $1, \tau$ and $1, \tau+1$ generate the same lattice, $\wp_\tau = \wp_{\tau+1}$. Since $\tau = \omega_2/\omega_1$, $-1/\tau = -\omega_1/\omega_2$ (with $\operatorname{Im}(-1/\tau) > 0$) corresponds essentially to

interchanging the periods, and indeed $\wp_{-1/\tau}(z) = \tau^2\wp_\tau(\tau z)$. So we consider the group of transformations of the upper half-plane generated by $\tau \mapsto \tau+1$ and $\tau \mapsto -1/\tau$, the **modular group**; quantities attached to $\wp_\tau$ should reflect these transformations.

2.1 Eisenstein series

The **Eisenstein series of order** k is

$$E_k(\tau) = \sum_{(n,m)\neq(0,0)} \frac{1}{(n+m\tau)^k} = \sum_{\omega\in\Lambda^*} \frac{1}{\omega^k},$$

for integers $k \geq 3$ and $\mathrm{Im}(\tau) > 0$.

Theorem 2.1

Eisenstein series have the following properties:
(i) The series $E_k(\tau)$ converges if $k \geq 3$, and is holomorphic in the upper half-plane.
(ii) $E_k(\tau) = 0$ if k is odd.
(iii) $E_k(\tau)$ satisfies the following transformation relations: $E_k(\tau+1) = E_k(\tau)$ and $E_k(\tau) = \tau^{-k}E_k(-1/\tau)$.

Property (iii) is the **modular character** of the Eisenstein series.

Proof. (i) By Lemma 1.5 and the remark after it, the series converges absolutely and uniformly in every half-plane $\mathrm{Im}(\tau) \geq \delta > 0$ for $k \geq 3$.

(ii) Replacing (n,m) by $(-n,-m)$ shows $E_k = -E_k$ for odd k.

(iii) $n + m(\tau+1) = (n+m) + m\tau$, and we may replace $n+m$ by n. Also $(n+m(-1/\tau))^k = \tau^{-k}(n\tau - m)^k$, and rearranging, replacing $(-m,n)$ by (n,m), gives the second relation. ∎

(Because of (ii), some authors define the Eisenstein series as $\sum 1/(n+m\tau)^{2k}$, possibly with a constant factor.)

Theorem 2.2

For z near 0, we have

$$\wp(z) = \frac{1}{z^2} + 3E_4z^2 + 5E_6z^4 + \cdots = \frac{1}{z^2} + \sum_{k=1}^{\infty}(2k+1)E_{2k+2}z^{2k}.$$

Proof. Replacing ω by $-\omega$, $\wp(z) = \frac{1}{z^2} + \sum_{\omega\in\Lambda^*}\left[\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2}\right]$. Differentiating the geometric series, $\frac{1}{(1-w)^2} = \sum_{\ell=0}^{\infty}(\ell+1)w^\ell$ for $|w|<1$, so for small z

$$\frac{1}{(z-\omega)^2} = \frac{1}{\omega^2}\sum_{\ell=0}^{\infty}(\ell+1)\Big(\frac{z}{\omega}\Big)^\ell = \frac{1}{\omega^2} + \sum_{\ell=1}^{\infty}(\ell+1)\frac{z^\ell}{\omega^{\ell+2}}.$$

Hence

$$\wp(z) = \frac{1}{z^2} + \sum_{\ell=1}^{\infty}(\ell+1)\Big(\sum_{\omega\in\Lambda^*}\frac{1}{\omega^{\ell+2}}\Big)z^\ell = \frac{1}{z^2} + \sum_{\ell=1}^{\infty}(\ell+1)E_{\ell+2}z^\ell = \frac{1}{z^2} + \sum_{k=1}^{\infty}(2k+1)E_{2k+2}z^{2k},$$

since $E_{\ell+2} = 0$ for odd ℓ. ∎

Hence near 0,

$$\wp'(z) = \frac{-2}{z^3} + 6E_4z + 20E_6z^3 + \cdots,$$

$$(\wp'(z))^2 = \frac{4}{z^6} - \frac{24E_4}{z^2} - 80E_6 + \cdots, \qquad (\wp(z))^3 = \frac{1}{z^6} + \frac{9E_4}{z^2} + 15E_6 + \cdots.$$

So $(\wp')^2 - 4\wp^3 + 60E_4\wp + 140E_6$ is holomorphic near 0 and vanishes there; being doubly periodic, it is constant (Theorem 1.2), hence 0:

Corollary 2.3

If $g_2 = 60E_4$ and $g_3 = 140E_6$, then $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$.

This is another version of Theorem 1.7, and expresses the symmetric functions of the e_j via Eisenstein series.

2.2 Eisenstein series and divisor functions

Consider the Fourier expansion of the periodic function $E_k(\tau)$; equivalently, the Laurent expansion in $z = e^{2\pi i\tau}$.

Lemma 2.4

If $k \geq 2$ and $\mathrm{Im}(\tau) > 0$, then

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\tau)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell \tau}.$$

Proof. One may apply Poisson summation to $f(z) = 1/(z+\tau)^k$. Alternatively, it suffices to treat $k = 2$, the other cases following by termwise differentiation. Differentiating $\sum_{n=-\infty}^{\infty} \frac{1}{n+\tau} = \pi \cot \pi\tau$ (Chapter 5) gives

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\tau)^2} = \frac{\pi^2}{\sin^2(\pi\tau)}.$$

With $w = e^{2\pi i\tau}$ ($|w| < 1$), $\sin^2 \pi\tau = -\frac{(e^{i\pi\tau} - e^{-i\pi\tau})^2}{4} = -\frac{(1-w)^2}{4w}$, so $\frac{\pi^2}{\sin^2 \pi\tau} = -4\pi^2 \frac{w}{(1-w)^2} = (-2\pi i)^2 \sum_{r=1}^{\infty} r w^r$, using $\sum_{r \geq 1} r w^r = \frac{w}{(1-w)^2}$. ∎

The **divisor function** $\sigma_\ell(r) = \sum_{d|r} d^\ell$ is the sum of the ℓth powers of the divisors of r.

Theorem 2.5

If $k \geq 4$ is even, and $\mathrm{Im}(\tau) > 0$, then

$$E_k(\tau) = 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i \tau r}.$$

Proof. $\sigma_{k-1}(r) \leq \sum_{d \leq r} d^{k-1} \leq r^k$; if $\mathrm{Im}(\tau) \geq t_0 > 0$, $|e^{2\pi i r\tau}| \leq e^{-2\pi r t_0}$, so the series converges absolutely, by comparison with $\sum r^k e^{-2\pi r t_0}$. Using the definition, k even, and the lemma with τ replaced by $m\tau$,

$$\begin{aligned} E_k(\tau) &= \sum_{n \neq 0} \frac{1}{n^k} + \sum_{m \neq 0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} = 2\zeta(k) + 2 \sum_{m>0} \sum_{n=-\infty}^{\infty} \frac{1}{(n+m\tau)^k} \\ &= 2\zeta(k) + 2\frac{(-2\pi i)^k}{(k-1)!} \sum_{m>0} \sum_{\ell=1}^{\infty} \ell^{k-1} e^{2\pi i \ell m \tau} = 2\zeta(k) + \frac{2(-1)^{k/2}(2\pi)^k}{(k-1)!} \sum_{r=1}^{\infty} \sigma_{k-1}(r) e^{2\pi i \tau r}, \end{aligned}$$

collecting the terms with $\ell m = r$. ∎

For the "forbidden" case $k=2$, $\sum_{(n,m)\neq(0,0)} 1/(n+m\tau)^2$ no longer converges absolutely, but define

$$F(\tau)=\sum_m\sum_n \frac{1}{(n+m\tau)^2}, \qquad (n,m)\neq(0,0),$$

summed in the indicated order (first n, then m).

Corollary 2.6

The double sum defining F converges in the indicated order. We have

$$F(\tau)=2\zeta(2)-8\pi^2\sum_{r=1}^{\infty}\sigma(r)e^{2\pi ir\tau},$$

where $\sigma(r)=\sum_{d|r} d$ is the sum of the divisors of r.

(The argument of Theorem 2.5 applies with $k=2$.) $F(-1/\tau)\tau^{-2}\neq F(\tau)$; equivalently, summing first in m and then in n gives a different value. Nevertheless this forbidden Eisenstein series plays a crucial role in the four-squares theorem (Chapter 10).

3 Selected exercises

Motivation. These exercises from Stein–Shakarchi (numbers kept) justify the normalization of the periods, add Abel's relation for zeros and poles, show why the exponent $r>2$ in Lemma 1.5 is sharp, and evaluate $\zeta(2)$, $\zeta(4)$ from the lattice sums.

Exercise 9.1 (Real ratio of periods)

Let f be meromorphic on $\mathbb{C}$ with non-zero periods ω_1,ω_2 such that $\omega_2/\omega_1\in\mathbb{R}$.
(a) If $\omega_2/\omega_1=p/q$ with p,q relatively prime integers, $q\geq 1$, show that f has the periods ω_1,ω_2 if and only if f has the single period $\omega_0=\omega_1/q$.
(b) If ω_2/ω_1 is irrational, show that f is constant.

Solution. If ω is a period, so is $k\omega$ for every $k\in\mathbb{Z}$ (induction on $k\geq 0$, and $f(z-\omega)=f((z-\omega)+\omega)$); if ω,ω' are periods, so is $\omega+\omega'$. Hence all integer combinations of periods are periods.

(a)

Step 1. If ω_0 is a period, then $\omega_1=q\omega_0$ and $\omega_2=\frac{p}{q}\omega_1=p\omega_0$ are integer multiples of it, hence periods.

Step 2. Conversely, since $\gcd(p,q)=1$ there are integers m,n with $mq+np=1$. Then

$$m\omega_1+n\omega_2=\omega_1\Big(m+n\frac{p}{q}\Big)=\frac{\omega_1}{q}(mq+np)=\omega_0,$$

an integer combination of ω_1,ω_2, hence a period.

(b) Let $\alpha=\omega_2/\omega_1\notin\mathbb{Q}$ and $D=\{m+n\alpha : m,n\in\mathbb{Z}\}$.

Step 1. (D is dense in $\mathbb{R}$.) Fix $N\geq 1$. Among the $N+1$ fractional parts of $0,\alpha,\dots,N\alpha$, two, say those of $j\alpha$ and $j'\alpha$ with $j\neq j'$, lie in the same interval $[i/N,(i+1)/N)$. Hence $\delta=(j-j')\alpha-\ell$ for a suitable $\ell\in\mathbb{Z}$ satisfies $|\delta|<1/N$, and $\delta\neq 0$ because α is irrational. All $k\delta$, $k\in\mathbb{Z}$, lie in D and are spaced by $|\delta|<1/N$, so every interval of length $1/N$ meets D. As N is arbitrary, D is dense.

Step 2. (Many small periods.) For $t=m+n\alpha\in D$, $t\omega_1=m\omega_1+n\omega_2$ is a period. By Step 1 choose $t_k\in D\setminus\{0\}$ with $t_k\to 0$.

Step 3. (Conclusion.) The set S of poles of f is discrete and closed, so $U = \mathbb{C} \setminus S$ is open and connected. Fix $z_0 \in U$. The points $z_k = z_0 + t_k\omega_1$ satisfy $z_k \neq z_0$, $z_k \to z_0$ and $f(z_k) = f(z_0)$ by Step 2. Thus the holomorphic function $f - f(z_0)$ on U has zeros accumulating at $z_0 \in U$, so $f \equiv f(z_0)$ on U. A function constant on a punctured neighborhood of a point of S cannot have a pole there, so $S = \varnothing$ and f is constant. ∎

Exercise 9.2 (Zeros minus poles)

Let f be an elliptic function with periods $1, \tau$ and lattice Λ, and let $a_1, \dots, a_r$ and $b_1, \dots, b_r$ be its zeros and poles in P_0, repeated according to multiplicity (there are equally many by Theorem 1.4). Show that

$$a_1 + \cdots + a_r - b_1 - \cdots - b_r \in \Lambda.$$

Solution.

Step 1. (Choice of a parallelogram.) The set Z of zeros and poles of f is countable. For $w \in Z$, the set of h with $w \in \partial(h + P_0)$ is $w - \partial P_0$, a union of four segments, which has area zero. A countable union of such sets has area zero, so there is h with no zero or pole on ∂P, where $P = h + P_0$.

Step 2. (Reduction to P.) By Proposition 1.1, each $z \in P_0$ is congruent to a unique $\phi(z) \in P$, and $\phi : P_0 \to P$ is a bijection with $\phi(z) - z \in \Lambda$. Since $f(z + \lambda) = f(z)$ for $\lambda \in \Lambda$, f has a zero (pole) of order k at z if and only if it has one at $\phi(z)$. Hence the zeros and poles in P are $a_i' = \phi(a_i)$ and $b_j' = \phi(b_j)$, and

$$\textstyle\sum_i a_i - \sum_j b_j = \sum_i a_i' - \sum_j b_j' - \lambda \qquad \text{for some } \lambda \in \Lambda.$$

So it suffices to show $\sum_i a_i' - \sum_j b_j' \in \Lambda$.

Step 3. (Residues.) Near a zero a of order k, $f'/f = k/(z-a) + (\text{holomorphic})$, so zf'/f has residue ka there; at a pole b of order k the residue is $-kb$. Elsewhere near $\overline{P}$ the function zf'/f is holomorphic. By the residue theorem,

$$\frac{1}{2\pi i}\int_{\partial P} z\frac{f'(z)}{f(z)}\,dz = \sum_i a_i' - \sum_j b_j'.$$

Step 4. (Pairing opposite sides.) Let $g = f'/f$, which has periods 1 and τ. The boundary runs $h \to h+1 \to h+1+\tau \to h+\tau \to h$. Substituting $z \mapsto z+\tau$ in the third side and $z \mapsto z+1$ in the second,

$$\int_h^{h+1} zg\,dz + \int_{h+1+\tau}^{h+\tau} zg\,dz = \int_h^{h+1} [z - (z+\tau)]g(z)\,dz = -\tau\int_h^{h+1} g(z)\,dz,$$
$$\int_{h+1}^{h+1+\tau} zg\,dz + \int_{h+\tau}^{h} zg\,dz = \int_h^{h+\tau} [(z+1) - z]g(z)\,dz = \int_h^{h+\tau} g(z)\,dz.$$

Step 5. (Integers.) The curve $\gamma(t) = f(h+t)$, $0 \le t \le 1$, is closed because $f(h+1) = f(h)$, and avoids 0. Hence

$$\int_h^{h+1} \frac{f'(z)}{f(z)}\,dz = \int_\gamma \frac{dw}{w} = 2\pi i\, n$$

with $n \in \mathbb{Z}$ the winding number of γ about 0. In the same way $\int_h^{h+\tau} g = 2\pi i\,\ell$ with $\ell \in \mathbb{Z}$.

Step 6. Combining Steps 3–5, $\sum_i a_i' - \sum_j b_j' = \ell - n\tau \in \Lambda$. ∎

Exercise 9.3 (The exponent 2 is critical)

(a) Show that $\sum_{\omega\in\Lambda^*}|\omega|^{-2}$ diverges for every $\tau\in\mathbb{H}$.
(b) Show that, as $R\to\infty$,

$$\sum_{1\le n^2+m^2\le R^2}\frac{1}{n^2+m^2}=2\pi\log R+O(1).$$

Solution.

(b) Write $\omega=(n,m)\in\mathbb{Z}^2$, $f(x)=|x|^{-2}$ on $\mathbb{R}^2\setminus\{0\}$, $c=\sqrt{2}/2$, and let Q_ω be the closed unit square centered at ω. Take $R\ge 4$.

Step 1. (Local comparison.) Let $|\omega|\ge 2$ and $x\in Q_\omega$. Every y on the segment $[\omega,x]\subset Q_\omega$ has $|y|\ge|\omega|-c\ge|\omega|/2$, and $|\nabla f(y)|=2|y|^{-3}\le 16|\omega|^{-3}$. The segment has length at most c, so by the mean value theorem

$$|f(x)-f(\omega)|\le 8\sqrt{2}\,|\omega|^{-3}.$$

Step 2. (Sum versus integral.) Split the sum into the six terms with $1\le|\omega|<2$, namely $(\pm1,0)$, $(0,\pm1)$, $(\pm1,\pm1)$, whose total is $4+4\cdot\frac12=6$, and the sum $S_1(R)$ over $2\le|\omega|\le R$. The squares have area 1 and disjoint interiors; let U_R be the union of Q_ω, $2\le|\omega|\le R$. By Step 1,

$$\Big|S_1(R)-\int_{U_R}f\,dx\Big|\le 8\sqrt{2}\sum_{|\omega|\ge2}|\omega|^{-3}=:C<\infty,$$

the series converging by Lemma 1.5 with $r=3$, $\tau=i$.

Step 3. (Annuli.) Let $A(a,b)=\{a\le|x|\le b\}$. If $2+c\le|x|\le R-c$, then x lies in some Q_ω, and $|x-\omega|\le c$ gives $2\le|\omega|\le R$; so $A(2+c,R-c)\subset U_R$. Conversely, $x\in Q_\omega$ with $2\le|\omega|\le R$ has $2-c\le|x|\le R+c$, so $U_R\subset A(2-c,R+c)$. In polar coordinates $\int_{A(a,b)}f=\int_0^{2\pi}\int_a^b r^{-1}\,dr\,d\theta=2\pi\log(b/a)$, and $f>0$, so

$$2\pi\log\frac{R-c}{2+c}\le\int_{U_R}f\,dx\le 2\pi\log\frac{R+c}{2-c}.$$

Step 4. (Conclusion.) For $R\ge4$, $|\log(1\pm c/R)|\le\log 2$, so both bounds equal $2\pi\log R+O(1)$. With Step 2 the full sum is $6+2\pi\log R+O(1)=2\pi\log R+O(1)$.

(a) With $\tau\in\mathbb{H}$ and $\omega=n+m\tau\neq0$,

$$|n+m\tau|\le|n|+|m||\tau|\le(1+|\tau|)(|n|+|m|)\le\sqrt{2}(1+|\tau|)(n^2+m^2)^{1/2}.$$

Hence $|\omega|^{-2}\ge\frac{1}{2(1+|\tau|)^2}(n^2+m^2)^{-1}$, and the partial sums over $n^2+m^2\le R^2$ are at least a fixed multiple of $2\pi\log R+O(1)\to\infty$ by (b). So the series diverges. ∎

Exercise 9.7 ($\zeta(2)$ and $\zeta(4)$ from lattice sums)

Using $\sum_{m\in\mathbb{Z}}(m+\tau)^{-2}=\pi^2/\sin^2(\pi\tau)$ at $\tau=1/2$, show

$$\sum_{m\ge1,\ m\text{ odd}}\frac{1}{m^2}=\frac{\pi^2}{8},\qquad\zeta(2)=\frac{\pi^2}{6},$$

and, using $\sum_{m\in\mathbb{Z}}(m+\tau)^{-4}$, show

$$\sum_{m\ge1,\ m\text{ odd}}\frac{1}{m^4}=\frac{\pi^4}{96},\qquad\zeta(4)=\frac{\pi^4}{90}.$$

Solution.

Step 1. (The identities on $\mathbb{C}\setminus\mathbb{Z}$.) By Chapter 5, $\pi\cot\pi z = \lim_N \sum_{|n|\le N} \frac{1}{z+n}$ for $z\notin\mathbb{Z}$, uniformly on compact subsets of $\mathbb{C}\setminus\mathbb{Z}$. Differentiating termwise (Weierstrass), as in the proof of Lemma 2.4,

$$\sum_{n\in\mathbb{Z}} \frac{1}{(z+n)^2} = \frac{\pi^2}{\sin^2\pi z} =: g(z), \qquad z\in\mathbb{C}\setminus\mathbb{Z},$$

and the left side converges absolutely and locally uniformly, so it may be differentiated twice more termwise:

$$\sum_{n\in\mathbb{Z}} \frac{6}{(z+n)^4} = g''(z).$$

Step 2. (Rewriting at $z=1/2$.) $(n+\frac12)^{-k} = 2^k(2n+1)^{-k}$, and as n runs over $\mathbb{Z}$, $2n+1$ runs over the odd integers, each odd $m\ge 1$ arising twice ($\pm m$). So for even k,

$$\sum_{n\in\mathbb{Z}} \frac{1}{(n+\frac12)^k} = 2^{k+1} \sum_{m\ge 1 \text{ odd}} \frac{1}{m^k}.$$

Step 3. ($k=2$.) $g(1/2)=\pi^2$, so $\sum_{m\text{ odd}} m^{-2} = \pi^2/8$. Splitting $\zeta(2)$ into odd and even terms, $\zeta(2) = \frac{\pi^2}{8} + \frac14\zeta(2)$, hence $\zeta(2)=\pi^2/6$.

Step 4. ($k=4$.) With $u=\pi z$,

$$g'(z) = -2\pi^3\csc^2 u\cot u, \qquad g''(z) = 2\pi^4\big(2\csc^2 u\cot^2 u + \csc^4 u\big).$$

At $z=1/2$, $\csc u = 1$ and $\cot u = 0$, so $g''(1/2) = 2\pi^4$ and $\sum_n (n+\frac12)^{-4} = \pi^4/3$. By Step 2, $\sum_{m\text{ odd}} m^{-4} = \frac{\pi^4}{3\cdot 32} = \frac{\pi^4}{96}$. Then $\zeta(4) = \frac{\pi^4}{96} + \frac{1}{16}\zeta(4)$, so $\zeta(4) = \frac{16}{15}\cdot\frac{\pi^4}{96} = \frac{\pi^4}{90}$. ∎